CSE 4403: Algorithms

Assignment Presentation on Complex Engineering Problem

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Selected Topic:

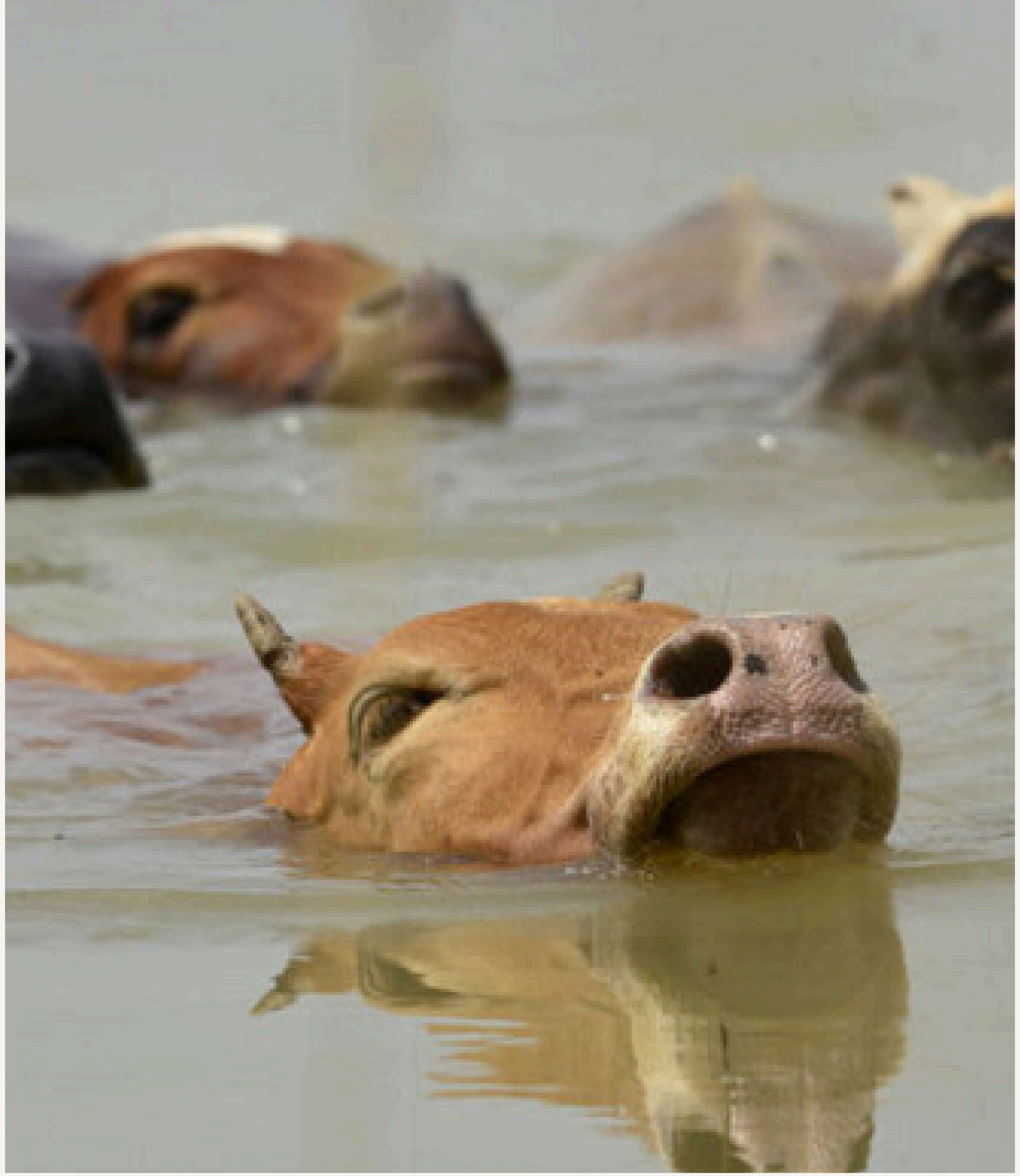
Flood Evacuation and Relief Distribution System

Introduction

The relevance and importance

Even after the government’s countless attempts and efforts, flooding remains to be a recurring and serious problem in many of urban areas of Bangladesh, especially large metropolitan areas like Dhaka and Chattogram. This year during the July flash floods in Chattogram, more than 866,000 people were affected, among which 43 were killed.

Thus a system where emergency rescue and relief operations are carried out as fast as possible are of paramount importance for a low-lying country, prone to flash floods like Bangladesh



Problems Addressed I

01 Checking the connectivity of a flood-affected area

02 Finding the best possible route from a source to a destination

03 Establishment of temporary communication or supply lines

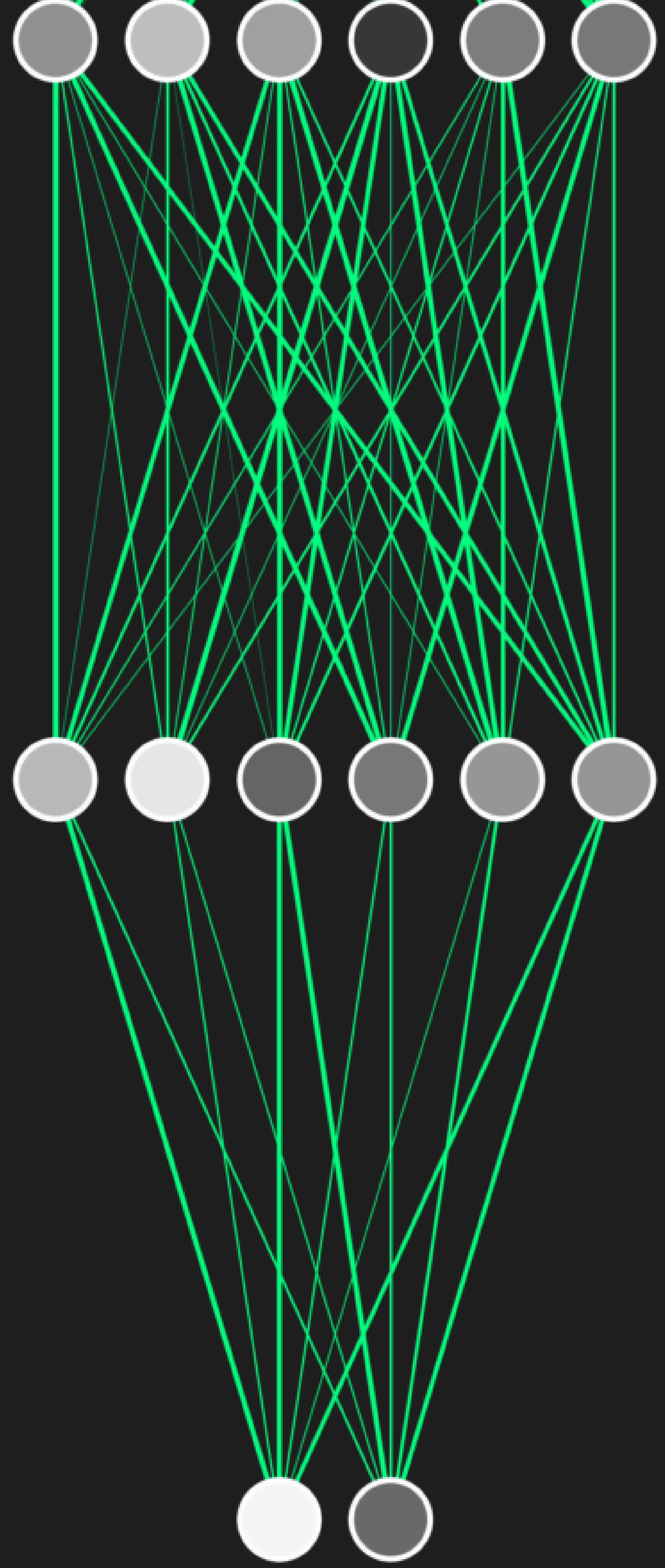
04 Calculating the best minimum costs and possible paths for every source-destination pair

Problems Addressed II

05 Ranking severity / importance of zones

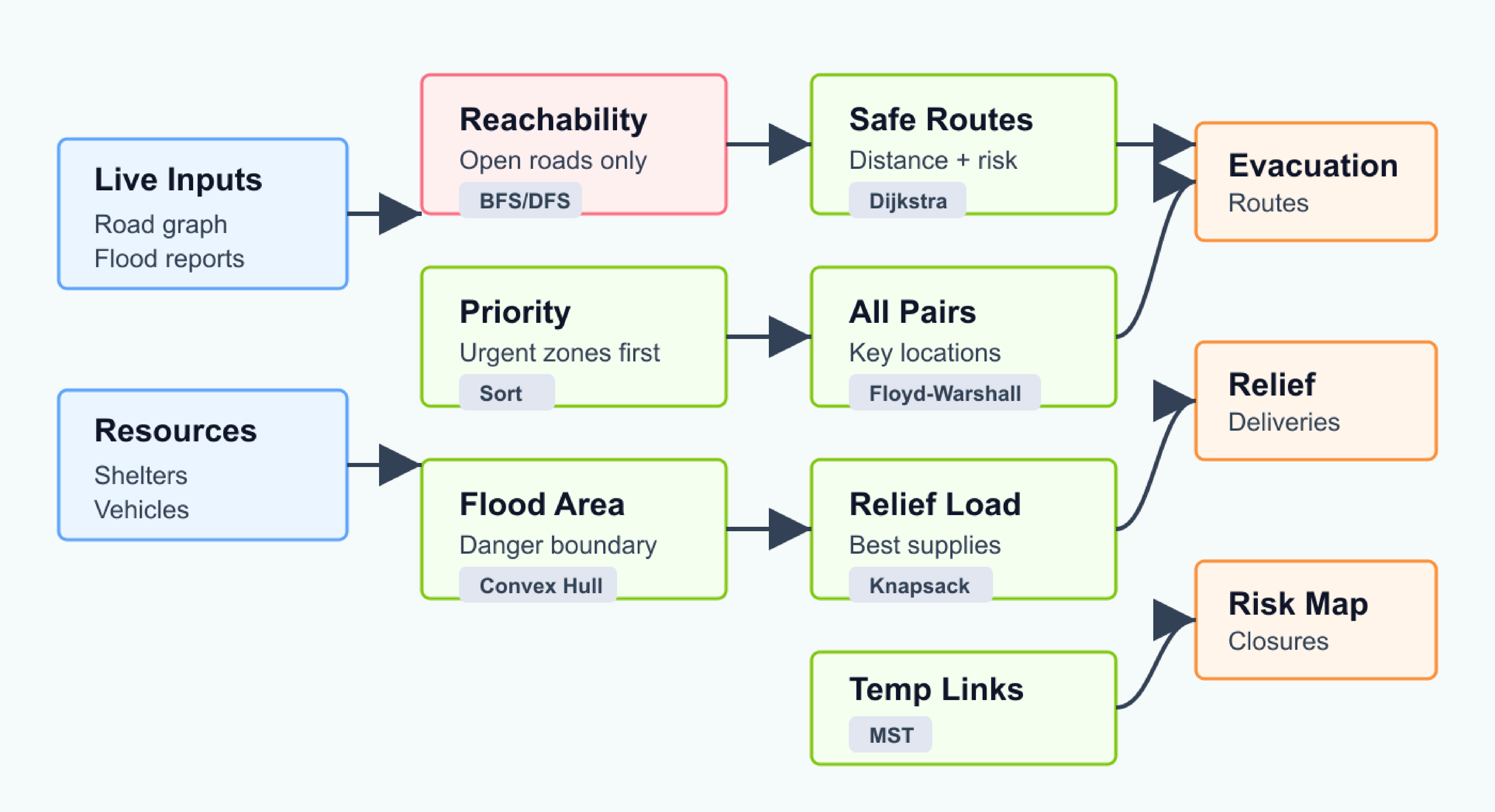
06 Mapping out the flood affected area and the relief centers that border it

07 Optimizing the value and weight of relief carried by relief teams.

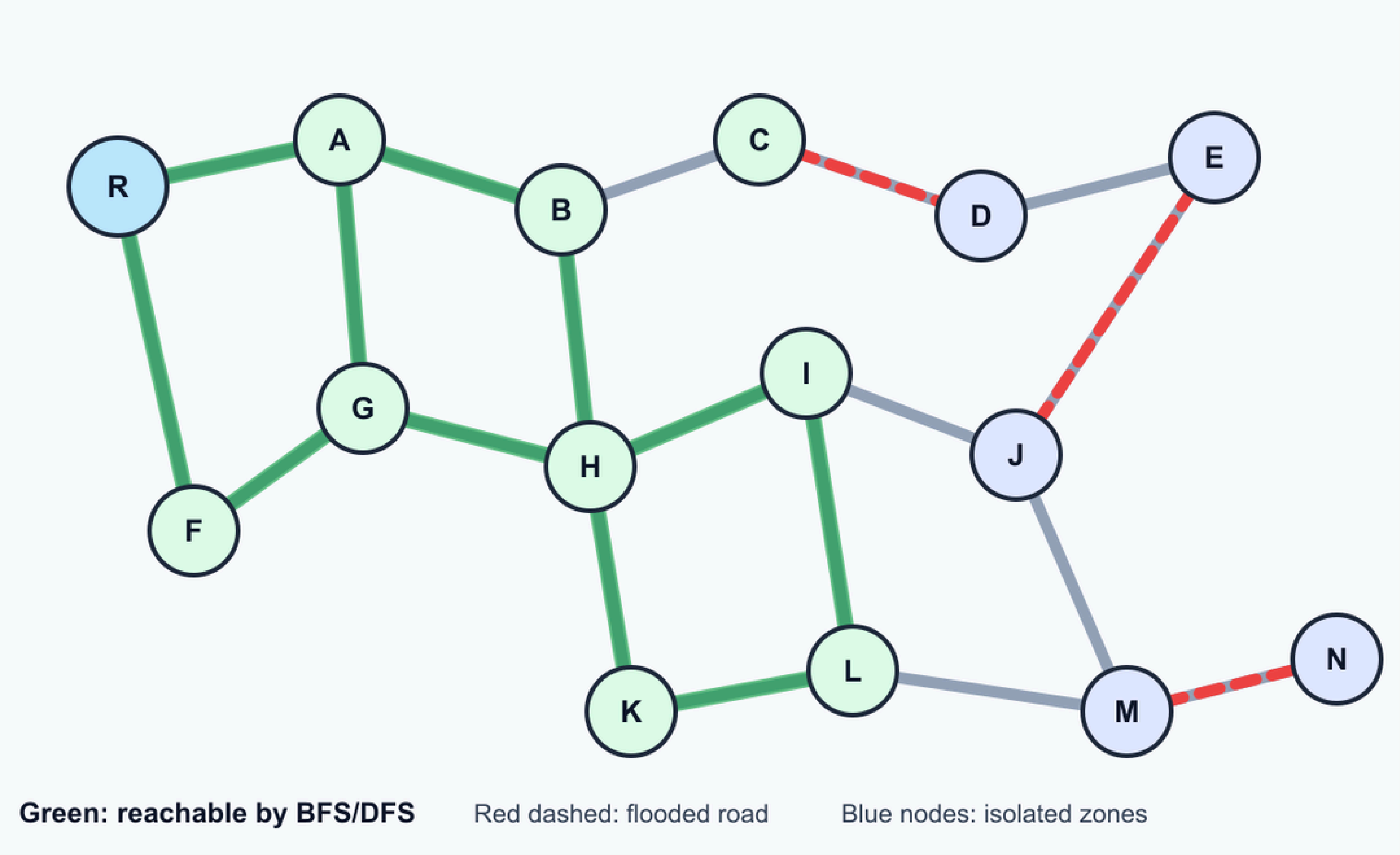


Applying Algorithms to Solve the Problems Addressed

System Pipeline



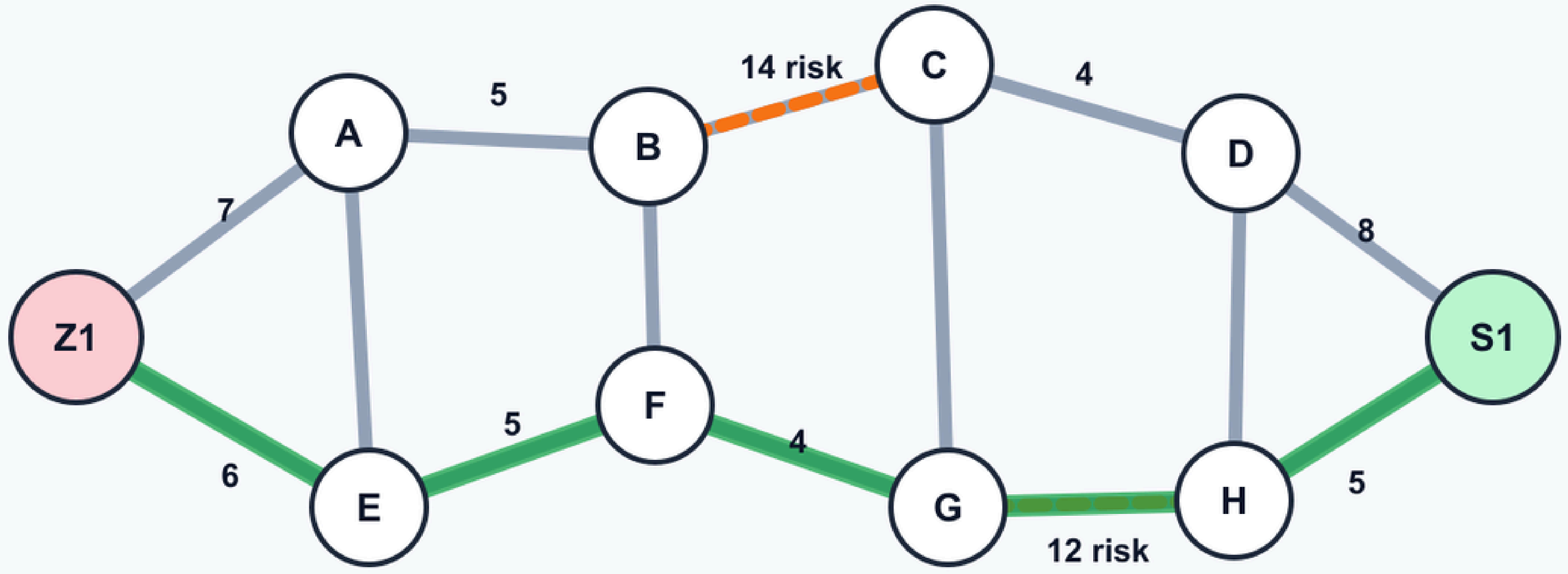
BFS and DFS



01

DFS can be used to check whether an affected area is reachable from rescue stations or shelters after flooded roads are removed from the graph. These algorithms can also identify disconnected components in the city network, helping authorities detect isolated zones that may require boats, helicopters, or special rescue teams

Dijkstra’s Algorithm



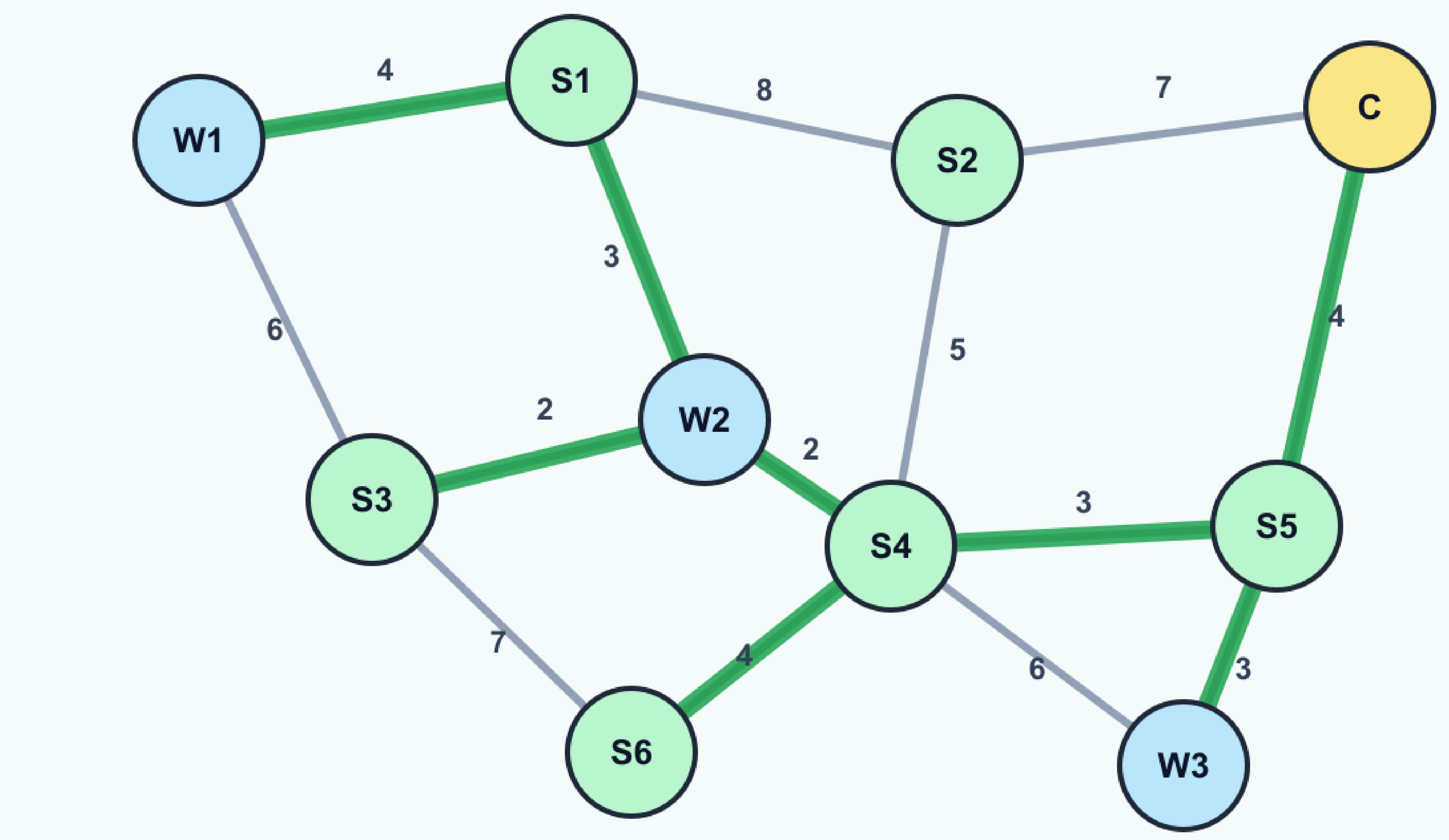
Selected path avoids high-risk roads

Shortest-looking route can be unsafe during flooding.

02

Dijkstra's algorithm can be used to find the shortest or fastest safe route from an affected zone to a shelter when all road weights are non-negative. The edge weight may represent travel time, distance, or a combined cost based on both distance and flood risk.

Prim’s / Kruskal’s Algorithm



Green links form the minimum spanning tree. Gray links are unused candidates.

03

If temporary communication lines, supply routes, or emergency service connections must be established among shelters and warehouses, MST algorithms can help choose a minimum-cost set of links that keeps all key locations connected.

Floyd-Warshall Algorithm

04

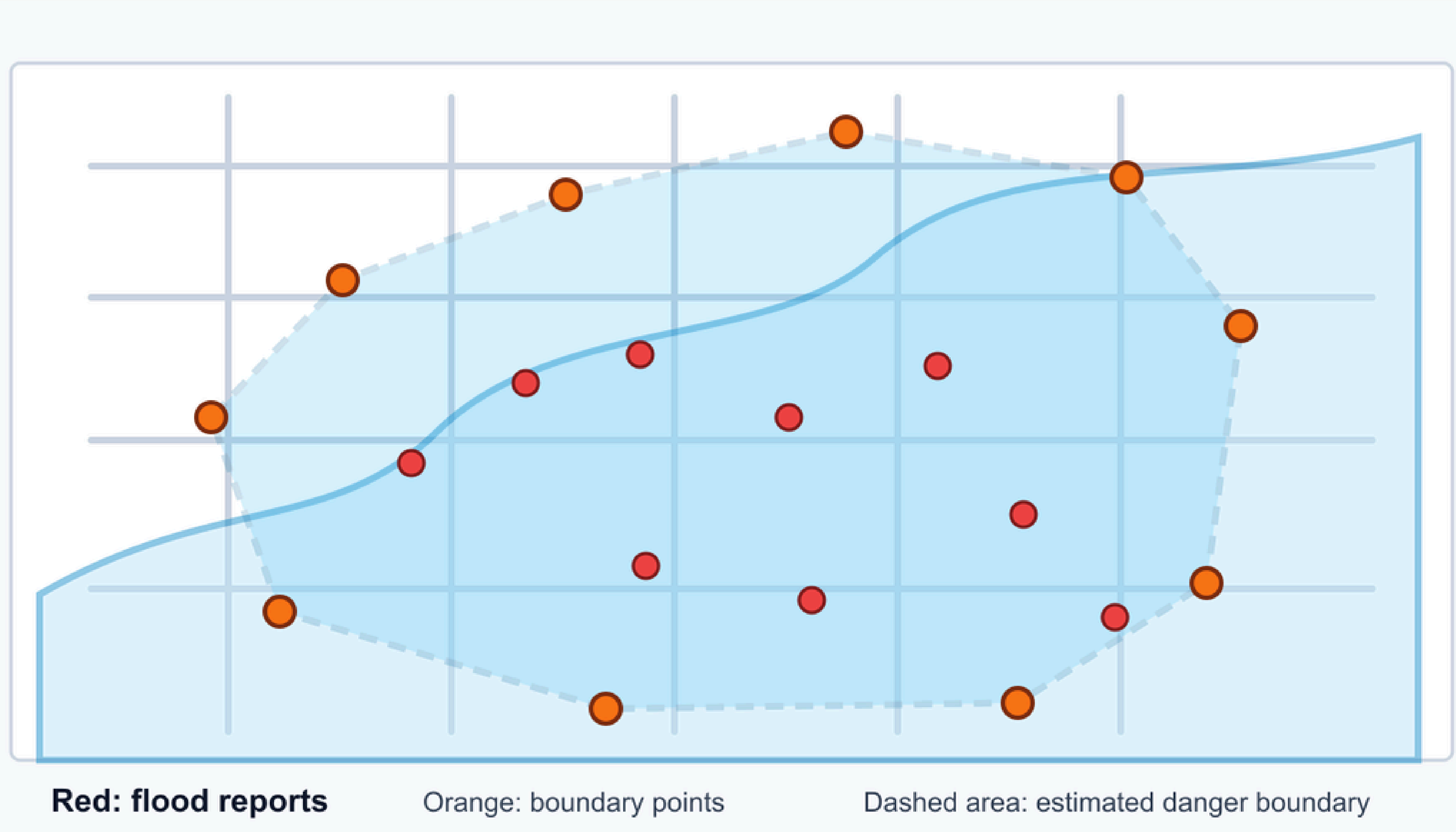
Floyd-Warshall can compute shortest paths between all important locations, such as shelters, warehouses, hospitals, and rescue stations. This is useful when authorities repeatedly need distance or travel-time information between many pairs of locations.

Sorting Algorithms

05

Sorting algorithms like merge sort or radix sort can rank affected zones by severity score, population size, estimated waiting time, or risk level. This helps emergency managers decide which areas should receive rescue teams or relief supplies first.

Convex Hull



06

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0/1 Knapsack

04

Relief vehicles have limited capacity, while each relief item has a weight and importance value. The 0/1 knapsack algorithm can select the best combination of supplies, such as medicine, food, and water, so that the total value is maximized without exceeding vehicle capacity.

CEP Critirias Met

CEP

01
Depth of Knowlege

The problem requires in-depth understanding of graph modeling, shortest path algorithms, graph traversal, minimum spanning tree, sorting, convex hull, and dynamic programming.

02
Conflicting Requirements

The system must balance shortest travel time, safest routes, shelter capacity, vehicle capacity, urgency of victims, and limited resources. A route that is shortest may not be safest, and a shelter that is nearest may already be full.

03
Depth of Analysis

There is no obvious single solution because evacuation planning, relief distribution, road availability, and shelter assignment influence each other.

CEP

04
Familiarity

Real-time flood evacuation and relief distribution is beyond standard classroom examples. It involves uncertain road conditions, changing flood boundaries, and emergency decisions under pressure.

06
Stakeholder Involvement

The problem involves citizens, rescue teams, shelter managers, hospitals, traffic authorities, city administrators, and relief organizations. These stakeholders may have different and sometimes conflicting priorities.

07
Inter-dependence

The problem contains many connected sub-problems like road reachability shelter assignment etc. A decision in one part affects the constraints and results of other parts.

Conclusion

- The optimized flood emergency evacuation and relief distribution system is a real-world engineering problem that strongly depends on algorithms studied in CSE 4403.
- Because the problem involves conflicting objectives, multiple stakeholders, and several interdependent components, it qualifies as a Complex Engineering Problem.
- It holds strong significance and relevance with the current situation of Bangladesh where each year many people lose their loved ones, livestock and properties to flooding due to lack of proper emergency and relief systems set up.

Thank You